# Arista Access Node QoS/Congestion Event Troubleshooting and Resolution Standard Operating Procedure

**Node:** BTM-ACC-19 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_6289

## Description

The Arista access node, situated at the BTM Layout Aggregation site, designated as BTM-ACC-19, plays a critical role in ensuring high-quality network connectivity. However, this node is susceptible to QoS/Congestion events, including microbursts, interface tail drops, and QoS queue starvation, which can severely impact network performance and reliability. These events often occur due to the complex interplay between various network protocols, hardware limitations, and the dynamic nature of network traffic. Specifically, on Arista equipment, the EOS (Extensible Operating System) manages the control and data planes, relying on a combination of hardware and software components to enforce QoS policies. When these policies are not properly aligned with the network's traffic patterns, or when hardware resources are insufficient to handle sudden spikes in traffic, congestion can occur, leading to the aforementioned failures. Understanding the root causes of these failures, which can range from misconfigured QoS policies to hardware limitations or unexpected traffic patterns, is crucial for effective troubleshooting and resolution. The Arista CLI provides a powerful toolset for diagnosing and addressing these issues, allowing for granular inspection of network behavior and precise adjustments to QoS configurations and other relevant settings.

## Pre-requisite Checks

- Verify site accessibility and safety for dispatching remote hands to BTM Layout Aggregation
- Check the status of redundant links to ensure no single point of failure
- Confirm the Arista node's power supply and cooling systems are operational

## Potential Root Causes

| Cause | Probability |
|---|---|
| Misconfigured QoS Policies | High |
| Hardware Resource Limitations | Medium |
| Unanticipated Traffic Patterns | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show qos interface e1, Expected Output: Detailed QoS configuration and statistics for interface e1, Description: This command isolates the current QoS configuration and traffic handling on the specified interface, helping to identify if the QoS policies are correctly applied and if there are any signs of congestion or packet
```

```
drops.

Command: show queue-monitor interface e1, Expected Output: Queue depth and drop statistics for interface e1, Description: This command provides insights into the real-time queue depths and drop statistics, which can indicate if the interface is experiencing congestion and if packets are being dropped due to queue overflow.
```

## Resolution Steps

Step 1: Step: Adjust QoS Policies, Command: qos policy-map QOS_POLICY_1 class QOS_CLASS_1 police rate 100m burst 10k, Description: This step involves adjusting the QoS policy to better match the current network traffic patterns, ensuring that critical traffic is prioritized and that no single traffic class can cause congestion. The specific command adjusts the policing rate and burst size for a particular class within a policy map.

Step 2: Step: Enable Priority Queue, Command: interface e1 qos trust dscp priority-queue out, Description: Enabling the priority queue on an interface ensures that critical traffic, marked with specific DSCP values, is prioritized and less likely to be dropped during congestion events. This command configures the interface to trust DSCP markings and to prioritize outgoing traffic accordingly.

## Post-Check Verification

```
Command: show qos interface e1, Expected Output: Updated QoS configuration and statistics showing no signs of congestion, Description: After applying the resolution steps, this command verifies that the QoS policies have been updated correctly and that there are no longer signs of congestion or packet drops on the interface.

Command: show queue-monitor interface e1, Expected Output: Queue depth and drop statistics indicating healthy queue operation, Description: This command checks the queue depths and drop statistics post-resolution, ensuring that the queues are operating within healthy parameters and that no packets are being dropped due to congestion.
```

## Roll-back Procedures

- Step: Revert QoS Policy Adjustments, Command: no qos policy-map QOS_POLICY_1, Description: If the applied resolution steps do not resolve the issue or introduce new problems, this step reverts the QoS policy adjustments to their previous state, ensuring that any changes made during troubleshooting are safely backed out.

- Step: Disable Priority Queue, Command: no interface e1 qos trust dscp priority-queue out, Description: This step disables the priority queue configuration that was enabled during the resolution process, returning the interface to its original configuration regarding priority queuing.